

FNP (Ferns and Petals)

Sales Data Analysis Report

Excel Dashboard & Business Intelligence Project

35,20,984 Total Revenue	1,000 Total Orders	3,520.98 Avg Order Value	5.53 days Avg Delivery Time
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Tools Used: Microsoft Excel | Pivot Tables | Dashboard | Data Analysis

1. Project Overview

Ferns and Petals (FNP) is a leading gifting company in India that specializes in delivering gifts across various occasions such as Diwali, Raksha Bandhan, Holi, Valentine's Day, Birthdays, and Anniversaries. This project involved analyzing the company's sales dataset using Microsoft Excel to uncover key insights related to revenue trends, customer behavior, and product performance.

Objective

To build an interactive Excel dashboard and answer 10 critical business questions that help FNP improve its sales strategy and enhance customer satisfaction.

Dataset Overview

Attribute	Details
Source	FNP Internal Sales Dataset
Period Covered	2023 – 2024
Total Records (Orders)	1,000
Key Fields	Order ID, Product, Category, Occasion, City, Revenue, Order Date, Delivery Date
Tools Used	Microsoft Excel (Pivot Tables, Charts, Slicers, Dashboard)

2. Key Performance Metrics (KPIs)

The following KPIs were calculated to provide a high-level view of FNP's business performance:

Total Revenue Generated	₹ 35,20,984
Total Number of Orders	1,000
Average Order Value (AOV)	₹ 3,520.98
Average Order-to-Delivery Time	5.53 Days
Number of Product Categories	7
Number of Cities Served	85
Occasions Covered	7 (All Occasions, Anniversary, Birthday, Diwali, Holi, Raksha Bandhan, Valentine's Day)

3. Business Questions & Analysis

Total Revenue

The total revenue generated across all orders and occasions in the dataset amounts to:

Grand Total Revenue: ₹ 35,20,984

Key Insights

- Revenue was tracked across 7 product categories and 7 occasion types.
- This figure was derived by summing all order revenues using Excel's SUM function within a Pivot Table.
- Colors category alone contributes ₹ 10,05,645 — nearly 28.6% of total revenue.

Average Order and Delivery Time

The time difference between Order Date and Delivery Date was calculated for each order using Excel's DATEDIF or subtraction formula. A new column 'diff_order_delivery' was derived.

Average Delivery Time	5.53 Days
Method Used	AVERAGE (diff_order_delivery column)
Insight	Orders are delivered within less than 6 days on average

Key Insights

- Average delivery time of 5.53 days indicates reasonably efficient logistics.
- This metric can be tracked per city or occasion to identify bottlenecks.

Monthly Sales Performance (2023)

Sales were grouped by month using a Pivot Table with Order Date as a row label (grouped by month). This shows how revenue fluctuates throughout the year.

Month (Order Date)	Sum of Revenue (₹)
Month 1 (January)	95,468
Month 2 (February)	7,04,509

Month (Order Date)	Sum of Revenue (₹)
Month 3 (March)	5,11,823
Month 4 (April)	1,40,393
Month 5 (May)	1,50,346
Month 6 (June)	1,57,913
Month 7 (July)	1,35,826
Month 8 (August)	7,37,389
Month 9 (September)	1,36,938
Month 10 (October)	1,51,619
Month 11 (November)	4,49,169
Month 12 (December)	1,49,591

Key Insights

- February is the highest revenue month (₹ 7,04,509) — driven by Valentine's Day.
- August sees a major spike (₹ 7,37,389) due to Raksha Bandhan and Independence Day gifting.
- November shows moderate recovery (₹ 4,49,169) likely tied to Diwali orders.
- January records the lowest revenue (₹ 95,468), indicating post-festive slowdown.
- Seasonal marketing campaigns should target Feb, Aug, and Nov as peak months.

Top Products by Revenue

Products were analyzed using a Pivot Table with Product Category as Row Labels and Sum of Revenue as Values. The top revenue-generating product categories are:

Product Category	Sum of Revenue (₹)	% of Total
Colors	10,05,645	28.6%
Soft Toys	7,40,831	21.0%
Sweets	7,33,842	20.8%
Cake	3,29,862	9.4%
Raksha Bandhan Items	2,97,372	8.4%
Plants	2,12,281	6.0%
Mugs	2,01,151	5.7%

Key Insights

- Colors is the top revenue category contributing 28.6% of total revenue.
- Soft Toys and Sweets together account for over 41% of total revenue.
- Mugs are the lowest revenue category — may need promotional boost.
- The top 3 categories (Colors, Soft Toys, Sweets) together generate ₹ 24,80,318 (70.4% of total).

. Customer Spending Analysis

Customer spending was evaluated by computing the average revenue per order across the entire dataset.

Average Revenue per Order	₹ 3,520.984
Total Orders Analyzed	1,000
Method	AVERAGE of Revenue column / Pivot Table Average
Highest Spending Occasion	All Occasions (₹ 5,86,176 total)
Lowest Spending Occasion	Valentine's Day (₹ 3,31,930 total)

Key Insights

- Average customer spends ₹ 3,520.98 per order, indicating a mid-premium gifting market.
- Customers purchasing for 'All Occasions' tend to have the highest collective spend.

- Personalized upselling strategies could increase AOV beyond ₹ 4,000.
- Gender split: Male customers contribute ₹ 17,90,223 vs Female's ₹ 17,30,761 — nearly equal.

Sales Performance by Top 5 Products

A focused analysis on the top 5 revenue-generating gift packages/products (not just categories):

Product Name	Revenue (₹)
Magnam Set	1,21,905
Quia Gift	1,14,476
Dolores Gift	1,06,624
Harum Pack	1,01,556
Deserunt Box	97,665

Grand Total of Top 5 Products: ₹ 5,42,226 (approx. 15.4% of total revenue)

Key Insights

- Magnam Set is the highest-selling individual product at ₹ 1,21,905.
- All top 5 products generate over ₹ 97,000 each — relatively balanced performance.
- These 5 products alone contribute 15.4% of total revenue, making them priority SKUs.
- Maintaining inventory and promotional focus on these products is critical during peak seasons.

Top 10 Cities by Number of Orders

City-wise order count was analyzed using a Pivot Table with City as Row Label and Count of Order_ID as values:

City	Number of Orders
Imphal	29
Dhanbad	28
Haridwar	24
Guntakal	20
Bidhannagar	21
Dibrugarh	21
Kavali	27
Bhatpara	18
Bilaspur	18
North Dumdum	19

Grand Total Orders from These Cities: 225

Key Insights
• Imphal leads with 29 orders, suggesting strong demand in the Northeast market.
• Dhanbad (28) and Kavali (27) are close runners-up.
• These Tier-2/3 cities show significant gifting demand — an untapped growth opportunity.
• FNP should expand delivery infrastructure and marketing in these high-order cities.
• Targeted regional campaigns could further drive order volume in these markets.

Order Quantity vs. Delivery Time

This analysis examines whether higher order quantities affect delivery times. The average delivery time was compared against order volumes across different time periods and categories.

Overall Average Delivery Time	5.53 Days
Analysis Method	Correlation (0.003478174)
Observation	Delivery time remains relatively stable across quantities

Key Insights
• Delivery time of ~5.53 days appears consistent regardless of order volume.
• No significant positive correlation observed between order quantity and delay.
• However, during peak months (Feb, Aug), slight delivery time increases are expected.
• FNP's logistics seem well-optimized; scaling operations for peak demand is recommended.
• Monitoring city-level delivery data could reveal localized bottlenecks.

Revenue Comparison Between Occasions

Revenue was segmented by occasion type using a Pivot Table with Occasion as Row Label and Sum of Revenue as values:

Occasion	Revenue (₹)	% Share
All Occasions	5,86,176	16.6%
Raksha Bandhan	6,31,585	17.9%
Holi	5,74,682	16.3%
Anniversary	6,74,634	19.2%
Diwali	3,13,783	8.9%
Birthday	4,08,194	11.6%
Valentine's Day	3,31,930	9.4%

Key Insights
• Anniversary generates the highest occasion-specific revenue at ₹ 6,74,634 (19.2%).
• Raksha Bandhan (₹ 6,31,585) and Holi (₹ 5,74,682) are strong revenue drivers.
• Diwali (₹ 3,13,783) underperforms relative to expectations — opportunity for growth.
• Valentine's Day revenue (₹ 3,31,930) can be boosted with targeted couple-focused campaigns.
• All Occasions collectively contribute ₹ 5,86,176 — consistent baseline demand exists.

Product Popularity by Occasion

Cross-analysis of product categories against occasions was performed using a multi-dimensional Pivot Table with slicers for occasion filtering (as visible in the dashboard).

Occasion	Most Popular Product	Revenue (₹)
Anniversary	Colors / Soft Toys	High
Birthday	Cake / Soft Toys	Moderate-High
Holi	Colors	Very High
Diwali	Sweets / Plants	Moderate
Raksha Bandhan	Raksha Bandhan Items / Sweets	High
Valentine's Day	Soft Toys / Cake	Moderate
All Occasions	Colors / Sweets	High

Key Insights

- Colors dominate Holi — the most logical and culturally aligned product.
- Soft Toys are popular across Anniversary and Valentine's Day.
- Sweets see strong performance during Diwali and Raksha Bandhan — traditional gifting culture.
- Cake is a consistent performer for Birthdays.
- FNP should curate occasion-specific bundles to maximize cross-sell opportunities.

Business Recommendations

- Prioritize Colors, Soft Toys, and Sweets in inventory planning — they drive 70%+ of revenue.
- Launch aggressive Valentine's Day and Diwali campaigns as these occasions underperform relative to their cultural potential.
- Expand logistics and marketing to Tier-2 cities like Imphal, Dhanbad, and Kavali which show high order volumes.
- Introduce occasion-specific product bundles (e.g., Holi Color Sets, Diwali Sweet Hampers) to boost average order value.
- Target February and August with pre-season stock-up and discounts to capitalize on peak demand.
- Investigate January's low revenue and run early-year promotions (New Year gifts, corporate gifting) to offset the slump.
- Maintain delivery SLA under 5 days to improve customer satisfaction ratings.
- Explore gender-personalized recommendations since male and female spending is nearly equal — both are valuable segments.

Conclusion

This data analysis project on FNP's sales dataset provided valuable insights into the company's revenue streams, customer behavior, and seasonal trends. Using Microsoft Excel's powerful Pivot Table and charting capabilities, a fully interactive dashboard was built that addresses 10 key business questions.

The analysis revealed that Colors, Soft Toys, and Sweets are the primary revenue drivers, with February and August being peak revenue months due to Valentine's Day and Raksha Bandhan respectively. Tier-2 cities demonstrate strong gifting demand, and occasion-specific purchasing patterns offer clear opportunities for targeted marketing strategies.

This project demonstrates proficiency in data cleaning, pivot analysis, dashboard creation, and data-driven business storytelling using Excel — skills directly applicable to roles in Data Analysis, Business Intelligence, and Sales Analytics.

Skills Demonstrated in This Project

Microsoft Excel | Pivot Tables | Data Visualization | Dashboard Design | Business Analysis | KPI Reporting